

# Rushil Shah

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## SKILLS

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**Languages:** Golang, Python, Java, C++, Ruby, Typescript, Javascript, Elixir, HTML5, CSS

**Technologies:** LLM, LangGraph, RabbitMQ, Kafka, React, Django, ElasticMQ, REST, InfluxDB, AWS, GraphQL

**Databases:** PostgreSQL, Elasticsearch, MongoDB, DynamoDB, Firebase, Cassandra, Redis

## PROFESSIONAL EXPERIENCE

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**Tesla** | Software Development Engineer

**Feb 2024 - Present**

- Converted a WebSocket-based **AI chatbot** integration to **HTTP streaming**, enhancing **scalability**, **reliability**, and **real-time performance** by reducing connection overhead and migrating to the **Orca infrastructure**.
- Designed **Autosweep** to clear out old data based on business requirements, saving **33%** of Tesla Advisors' time via **Goroutines** and **Cron Jobs**, configurable via **SQL** query without production releases for future changes.
- Engineered real-time email notification feature by programming cron jobs to poll data at regular intervals, and asynchronously send the email payload to the necessary **RabbitMQ** queues for message transfer.
- Accomplished a **40%** reduction in write errors, due to version conflict while writing documents in ElasticSearch by implementing a retry mechanism with error handling, random jitters, and async programming, using **Goroutines**
- Enabled storing **Database transactions** in request context to avoid creating multiple transactions for every database call using the **Golang Context** package, improving the CPU consumption by 20% and drastically reducing DB load

**Fetch** | Backend Engineer

**Nov 2023 - Feb 2024**

- Scaling microservices that scan millions of user receipts, by implementing microservices and pipelines in **Golang** to check validations, item expiry, normalization, etc, and storing data in the **Snowflake** data lake
- Lead migration of **314 GB** of data from **AWS ElastiCache** to a compact version of **DynamoDB**, optimizing CPU utilization from **95%** to **32%** through effective fallback and feature flag implementation

**Influxdata** | Software Engineering Intern ([General Marketplace Simulator](#))

**May 2022 - Aug 2022**

- Designed a **Notification service** to simulate the **AWS Marketplace** for testing the onboarding of **70%** of company users, using functional programming in **Elixir Phoenix**, to send messages to local queues in **Docker**
- Conceptualized and implemented the architecture to duplicate **AWS SNS** and **SQS** services locally in **Elixir**, by sending notifications at run time and polling message queues at regular intervals, locally using **ElasticMQ**

**Shipmnts** | Full Stack Software Engineer

**Dec 2020 - Jun 2021**

- Improved scalability by **30%** in a Subscription service that delivers tracking data in real-time, by using a **queueing architecture** in **Ruby on Rails** to delegate large asynchronous tasks to different queues for information retrieval
- Achieved a **50%** reduction in API consumption in a subscription service by programming **Cron Jobs** in **JavaScript** to fetch and update data asynchronously, and adjusting cron time intervals at runtime based on previous tracking data

**Taasha Technologies** | Full Stack Software Engineer

**Jun 2020 - Sep 2020**

- Developed an E-commerce application for over **20000** products, using **Java Spring**, **React**, and **RESTful APIs** in a microservice architecture, which included services like catalog, user authentication, order processing, etc
- Led the optimization of search filters and latency reduction efforts, resulting in a **30%** improvement in performance using **PostgreSQL** and **Redis** by implementing a hierarchical data structure and writing efficient **SQL** queries

## EDUCATION

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**Master of Science in Software Engineering** | San Jose State University, California - **GPA: 3.81/4**

**Aug 2023**

## PROJECTS

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**Vini: AI Driven chatbot** ([Vini](#)) Python | Grok | Vector DB | FastAPI

**Feb 2023 - May 2023**

- Designed and implemented a personalized chatbot using **LLMs** and vector search to deliver context-aware responses about my education, work experience, and skills, ensuring accurate and dynamic information retrieval.
- Optimized embeddings and retrieval flow, improving response relevance and query accuracy by **40%**, demonstrating expertise in **AI-driven** personalization and natural language understanding.